

Data Analysis using SQL (Amazon Brazil)



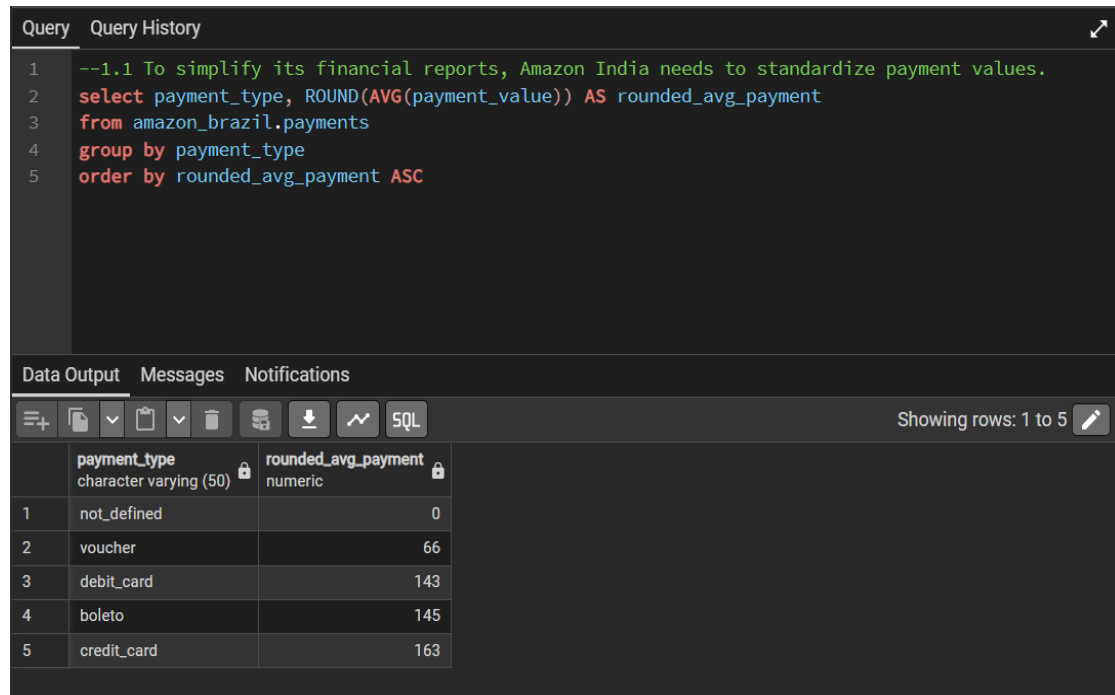
Introduction:

- [Amazon](#), a global leader in e-commerce, has achieved significant success in markets like the U.S., Europe, and Asia. In Brazil, Amazon connects small and medium businesses with millions of customers, becoming a key player. Given the similarities between Brazil and India—such as large populations and diverse consumer bases—there's an opportunity to replicate success in India.
- The primary goal is to identify actionable insights, customer behaviour patterns, and market trends that can be strategically leveraged to accelerate growth and enhance customer experience for Amazon in the Indian market.

Analysis 1

1. To simplify its financial reports, Amazon India needs to standardize payment values.

SQL query snippet:



```
1 --1.1 To simplify its financial reports, Amazon India needs to standardize payment values.
2 select payment_type, ROUND(AVG(payment_value)) AS rounded_avg_payment
3 from amazon_brazil.payments
4 group by payment_type
5 order by rounded_avg_payment ASC
```

	payment_type character varying (50)	rounded_avg_payment numeric
1	not_defined	0
2	voucher	66
3	debit_card	143
4	boleto	145
5	credit_card	163

Analysis: This query reveals the typical transaction size associated with each payment method. For instance, credit cards in Brazil likely have a high average payment value, indicating they are used for significant purchases. Vouchers, conversely, have a very low average value, suggesting they are used for small, supplementary payments or promotions.

Recommendation for India: Recognize the distinct roles of different payment methods. Promote UPI as the primary, frictionless option for low-to-mid value daily transactions. For high-value items, heavily feature EMI (Equated Monthly Installment) options on credit cards, debit cards, and "Buy Now, Pay Later" (BNPL) services at the point of checkout to make expensive products more accessible and improve conversion rates.

2. To refine its payment strategy, Amazon India wants to know the distribution of orders by payment type.

SQL query snippet:

```
7 --1.2 To refine its payment strategy, Amazon India wants to know the distribution of orders by payment type.
8 select payment_type,
9 round((count(*) * 100.0 / (select count(*) from amazon_brazil.payments)), 1) as percentage_orders
10 from amazon_brazil.payments
11 group by payment_type
12 order by percentage_orders desc
13
```

Data Output			Messages	Notifications
			Showing rows: 1 to 5	Page No: 1
	payment_type character varying (50)	percentage_orders numeric		
1	credit_card	73.9		
2	boleto	19.0		
3	voucher	5.6		
4	debit_card	1.5		
5	not_defined	0.0		

Analysis: The data shows the market share of each payment method. In Brazil, credit cards are the dominant form of payment, accounting for the vast majority of transactions. This indicates a mature market with high consumer trust in digital card payments.

Recommendation for India: Amazon India must acknowledge that UPI is the dominant payment method. It should be the default and most prominent option during checkout. Additionally, while declining, Cash on Delivery (CoD) still holds a significant share in Tier-2 and Tier-3 cities and should be retained as an option to cater to that customer base.

3. Amazon India seeks to create targeted promotions for products within specific price ranges.

SQL query snippet:

```
13
14 --1.3 Amazon India seeks to create targeted promotions for products within specific price ranges.
15 SELECT p.product_id, oi.price FROM amazon_brazil.products p
16 JOIN
17     amazon_brazil.order_items oi ON p.product_id = oi.product_id
18 WHERE
19     oi.price BETWEEN 100 AND 500
20     AND p.product_category_name ILIKE '%Smart%'
21 ORDER BY oi.price DESC;
22
```

Data Output Messages Notifications

Showing rows: 1 to 34

Page No:

	product_id character varying (255)	price numeric (10,2)
1	1df1a2df8ad2b9d3aa49fd851e314...	439.99
2	7debe59b10825e89c1cbcc8b190c...	349.99
3	ca86b9fe16e12de698c955aedff0ae...	349.00
4	ca86b9fe16e12de698c955aedff0ae...	349.00
5	0e52955ca8143bd179b311cc454a...	335.00
6	7aeaa8f3e592e380c420e8910a717...	329.90
7	7aeaa8f3e592e380c420e8910a717...	329.90
8	7aeaa8f3e592e380c420e8910a717...	329.90
9	7aeaa8f3e592e380c420e8910a717...	329.90
10	7aeaa8f3e592e380c420e8910a717...	329.90
11	7aeaa8f3e592e380c420e8910a717...	329.90
12	d1b571cd58267d8cac8b2afd6e288...	299.90

Total rows: 34 Query complete 00:00:00.175

Analysis: This query identifies a specific, popular product segment ('Smart' devices) within a key mid-range price bracket. This confirms the existence of a high-potential audience for targeted marketing campaigns.

Recommendation for India: Replicate this targeting strategy by identifying locally relevant, high-traction keywords (e.g., "AI-powered," "5G ready," "fast charging") within popular price segments (e.g., smartphones under ₹20,000). Create dedicated ad campaigns and landing pages for these segments, especially during major sales events like Diwali and Prime Day.

4. To identify seasonal sales patterns, Amazon India needs to focus on the most successful months.

SQL query snippet:

```
25
26 --1.4 To identify seasonal sales patterns, Amazon India needs to focus on the most successful months.
27 select to_char(o.order_purchase_timestamp, 'Month') as month,
28        round(sum(p.payment_value)) as total_sales
29 from amazon_brazil.orders o
30 join
31      amazon_brazil.payments p on o.order_id = p.order_id
32 group by month
33 order by total_sales desc
34 limit 3
35
36
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No:

	month text	total_sales numeric
1	May	1746901
2	August	1696822
3	July	1658924

Analysis: Sales are not evenly distributed; distinct months show significant revenue peaks. In Brazil, these peaks are likely tied to major commercial holidays like Black Friday and Christmas, indicating that customers are conditioned to spend more during these periods.

Recommendation for India: Amazon India should build its core marketing and inventory calendar around India's key consumption periods. This includes major festivals like Diwali, Dussehra, and Eid, as well as national events like Republic Day and Independence Day, which have become major online shopping seasons.

5. Amazon India is interested in product categories with significant price variations.

SQL query snippet:

```
37 --1.5 Amazon India is interested in product categories with significant price variations.
38 select p.product_category_name , (max(oi.price) - min(oi.price)) as price_difference
39 from amazon_brazil.products p
40 join
41     amazon_brazil.order_items oi on p.product_id = oi.product_id
42 group by
43     p.product_category_name
44 having
45     (max(oi.price) - min(oi.price)) > 500
46 order by price_difference desc
47
48
```

	product_category_name character varying (255)	price_difference numeric
1	utilidades_domesticas	6731.94
2	pcs	6694.50
3	artes	6495.50
4	eletroportateis	4792.50
5	instrumentos_musicais	4394.97
6	consoles_games	4094.81
7	esporte_lazer	4054.50
8	relogios_presentes	3990.91
9	[null]	3977.00
10	ferramentas_jardim	3923.65
11	bebes	3895.46
12	informatica_acessorios	3696.09

Total rows: 57 Query complete 00:00:00.188

Analysis: Categories such as electronics, home appliances, and watches show a large difference between their lowest and highest-priced items. This means they successfully cater to a wide spectrum of customers, from budget-conscious shoppers to premium buyers.

Recommendation for India: For these high-variance categories, enhance the user experience by implementing robust "Shop by Price" filters. Create curated storefronts like "Best under ₹1000" or a "Luxury & Premium Collection" to help different customer personas navigate the vast selection and find products relevant to their budget.

6. To enhance the customer experience, Amazon India wants to find which payment types have the most consistent transaction amounts.

SQL query snippet:

```
48
49
50 --1.6 To enhance the customer experience, Amazon India wants to find which payment types have the most cons
51 select payment_type, STDDEV(payment_value) AS std_deviation
52   from amazon_brazil.payments
53  group by
54   payment_type
55  order by
56   std_deviation asc
57
58
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1

	payment_type character varying (50)	std_deviation numeric
1	not_defined	0
2	voucher	115.519185430458
3	boleto	213.581061475527
4	credit_card	222.119310741208
5	debit_card	245.793401040647

Analysis: The standard deviation reveals the predictability of transaction sizes. Vouchers show very low deviation, meaning they are consistently used for similar, small amounts. Credit cards exhibit high deviation, as they are used for everything from minor purchases to very large ones.

Recommendation for India: Use this insight for financial forecasting and risk management. Revenue from high-volume, low-deviation methods like UPI is highly predictable. For high-deviation methods like credit cards, which cover a wider range of values, fraud detection algorithms should be more rigorously applied.

7. Amazon India wants to identify products that may have incomplete name in order to fix it from their end.

SQL query snippet:

```
58
59 --1.7 Amazon India wants to identify products that may have incomplete name in order to fix it from their e
60 select product_id, product_category_name
61 from amazon_brazil.products
62 where
63 product_category_name is null
64 or length(product_category_name) = 1
65
```

Data Output Messages Notifications		
Showing rows: 1 to 614 Page No: 1		
	product_id [PK] character varying (255)	product_category_name character varying (255)
1	a41e356c76fab66334f36de622ecbd...	[null]
2	d8dee61c2034d6d075997acef1870...	[null]
3	56139431d72cd51f19eb9f7dae4d16...	[null]
4	46b48281eb6d663ced748f324108c...	[null]
5	5fb61f482620cb672f5e586bb132ea...	[null]
6	e10758160da97891c2fdbc35f0f03...	[null]
7	39e3b9b12cd0bf8ee681bbc1c130fe...	[null]
8	794de06c32a626a5692ff50e4985d3...	[null]
9	7af3e2da474486a3519b0cba9dea8...	[null]
10	629beb8e7317703dcc5f35b5463fd2...	[null]
Total rows: 614 Query complete 00:00:00.139		

Analysis: The query identifies a data quality issue where some products lack proper categorization. This is a critical problem as it directly impacts product discoverability through search and browsing, leading to lost sales.

Recommendation for India: Implement a "Seller Health Dashboard" that automatically flags products with missing or poor-quality data. To incentivize sellers to maintain high standards, products with incomplete information could be temporarily down-ranked in search results until the issues are resolved.

Analysis 2

1. Amazon India wants to understand which payment types are most popular across different order value segments (e.g., low, medium, high).

SQL query snippet:

```
67
68 --2.1 Amazon India wants to understand which payment types are most popular across different order value se
69 select
70     case
71         when payment_value < 200 then 'Low Value (<200)'
72         when payment_value between 200 and 1000 then 'Medium Value (200-1000)'
73         else 'High Value (>1000)'
74     end as order_value_segment,
75     payment_type, count(*) as count
76 from
77     amazon_brazil.payments
78 group by
79     order_value_segment,
80     payment_type
81 order by
82     order_value_segment,
83     count desc
84
```

	order_value_segment text	payment_type character varying (50)	count bigint
1	High Value (>1000)	credit_card	944
2	High Value (>1000)	boleto	178
3	High Value (>1000)	debit_card	15
4	High Value (>1000)	voucher	13
5	Low Value (<200)	credit_card	60548
6	Low Value (<200)	boleto	16444
7	Low Value (<200)	voucher	5476
8	Low Value (<200)	debit_card	1287

Total rows: 13 Query complete 00:00:00.146

Analysis: This query shows that while credit cards are popular across all segments, their dominance increases with the order value. Lower-cost methods like boleto are strong for low and medium-value purchases but are less trusted or convenient for high-value transactions.

Recommendation for India: Tailor the checkout experience based on the cart value. For high-value carts, dynamically highlight EMI offers and card-specific discounts. For low-value carts, prioritize speed and convenience by making UPI and digital wallets the most prominent one-click payment options.

2. Amazon India wants to analyse the price range and average price for each product category.

SQL query snippet:

```
85
86 --2.2 Amazon India wants to analyse the price range and average price for each product category.
87 select
88     p.product_category_name,
89     min(oi.price) as min_price,
90     max(oi.price) as max_price,
91     avg(oi.price) as avg_price
92 from
93     amazon_brazil.products p
94 join
95     amazon_brazil.order_items oi on p.product_id = oi.product_id
96 where
97     p.product_category_name is not null
98 group by
99     p.product_category_name
100 order by
101     avg_price desc
102
103
```

	product_category_name character varying (255)	min_price numeric	max_price numeric	avg_price numeric
1	pcs	34.50	6729.00	1098.3405418719211823
2	portateis_casa_forno_e_cafe	10.19	2899.00	624.2856578947368421
3	eletrodomesticos_2	13.90	2350.00	476.1249579831932773
4	agro_industria_e_comercio	12.99	2990.00	341.6610426540284360
5	instrumentos_musicais	4.90	4399.87	281.6160000000000000
6	eletroportateis	6.50	4799.00	280.7784683357879234
7	portateis_cozinha_e_preparadores_de_aliment...	17.42	1099.00	264.5686666666666667
8	telefonos_fixos	6.00	1790.00	225.6931818181818182
Total rows: 78		Query complete 00:00:00.185		

Analysis: The min, max, and average price data reveals the market positioning of each category. Categories like watches_gifts may have a high average price, signaling a premium or luxury market. Others like stationery have a low average price, suggesting a volume-driven, everyday-essentials market.

Recommendation for India: Develop category-specific business strategies. For high-average-price categories, focus on building trust through brand partnerships, authenticity guarantees ("Brand Authorized Seller" tags), and premium customer service. For low-average-price categories, focus on driving volume through combo deals, bulk discounts, and "Subscribe & Save" models.

3. Amazon India wants to identify the customers who have placed multiple orders over time.

SQL query snippet:

```
103
104 --2.3 Amazon India wants to identify the customers who have placed multiple orders over time.
105 select c.customer_unique_id, count(o.order_id) as total_orders
106     from
107     amazon_brazil.customers c
108  join
109  amazon_brazil.orders o on c.customer_id = o.customer_id
110  group by
111  c.customer_unique_id
112  having
113  count(o.order_id) > 1
114  order by
115  total_orders DESC
116
117
```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1

	customer_unique_id character varying (255)	totalOrders bigint
1	a91e80fbe80ddc07de66a5cf927029...	16
2	a6168cd79131e64acef92e3c74d6cc...	16
3	363f980585bf04c1a88fdb986011c5...	16
4	cbd0350d4ccba9772e8e768d4a4a5...	16
5	417b909c0962b2610f1cfeb1c14789...	16
6	5f94af52aef02c968a2e0f01f430864e	16
7	1b6d29725255a77667a8c639eeb4c...	16
8	e4bbcc533fdf3917c56dea2c43bf20...	16

Total rows: 3140 Query complete 00:00:00.904

Analysis: The data confirms that a significant portion of business is driven by a smaller group of repeat customers. These customers are the most valuable asset, and identifying them is the crucial first step toward building a loyalty-driven business model.

Recommendation for India: Tag these multi-purchase customers within the CRM system. Prioritize them for marketing efforts by sending exclusive "Just for You" offers, early access to sales, and personalized product recommendations to encourage their next purchase and solidify their loyalty.

4. Amazon India wants to categorize customers into different types ('New – order qty. = 1' ; 'Returning' –order qty. 2 to 4; 'Loyal' – order qty. >4) based on their purchase history.

SQL query snippet:

```
118 --2.4 Amazon India wants to categorize customers into different types ('New – order qty. = 1' ; 'Returnin
119 with customer_order_counts as (
120     select c.customer_unique_id,
121           count(o.order_id) as total_orders
122     from
123         amazon_brazil.customers c
124     join
125         amazon_brazil.orders o on c.customer_id = o.customer_id
126     group by
127         c.customer_unique_id
128 )
129 select customer_unique_id,
130        case
131            when total_orders = 1 then 'New'
132            when total_orders between 2 and 4 then 'Returning'
133            else 'Loyal'
134        end as customer_type
135 from
136     customer_order_counts
137 order by
138     total_orders desc |
```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1

	customer_unique_id character varying (255)	customer_type text
1	6af40347f5dd7bdd65437a35e1b2f...	Loyal
2	7f4f709af2fd8fea44aacd30bca462...	Loyal
3	417b909c0962b2610f1cfef1c1478...	Loyal
4	75f15790b1852b42b1dbf645d98ff...	Loyal
5	0fdc0d21e1983e8af4d399e17671f...	Loyal
6	363f980585bf04c1a88fdb986011c...	Loyal
7	5f94af52aef02c968a2e0f01f43086...	Loyal

Total rows: 95077 Query complete 00:00:01.028

Analysis: This query quantifies the customer base into clear lifecycle stages: 'New', 'Returning', and 'Loyal'. This provides a snapshot of business health, showing the balance between new customer acquisition and the retention of existing customers.

Recommendation for India: Automate marketing communications based on these segments. Implement distinct "customer journeys": a welcome and onboarding series for 'New' customers, a re-engagement campaign for 'Returning' customers, and a VIP rewards program for 'Loyal' customers.

5. Amazon India wants to know which product categories generate the most revenue.

SQL query snippet:

```
141 --2.5 Amazon India wants to know which product categories generate the most revenue.
142 select p.product_category_name, sum(oi.price) as total_revenue
143     from amazon_brazil.products p
144  join
145     amazon_brazil.order_items oi on p.product_id = oi.product_id
146  where
147     p.product_category_name is not null
148  group by
149     p.product_category_name
150  order by
151     total_revenue desc
152  limit 5
153
```

Data Output Messages Notifications

Showing rows: 1 to 5

	product_category_name character varying (255)	total_revenue numeric
1	beleza_saude	1257865.34
2	relogios_presentes	1203060.32
3	cama_mesa_banho	1032268.59
4	esporte_lazer	985881.10
5	informatica_acessorios	910605.07

Total rows: 5 Query complete 00:00:00.253

Analysis: The Pareto principle is evident here: a small number of key categories are responsible for a large majority of the total revenue. These categories are the financial backbone of the business.

Recommendation for India: Designate these as "Key Value Categories" (KVCs). Allocate disproportionately more resources to them, including dedicated category managers, higher ad spend, enhanced seller support, and a more resilient supply chain to prevent stockouts at all costs.

Analysis 3

1. The marketing team wants to compare the total sales between different seasons.

SQL query snippet:

```
156 --3.1 The marketing team wants to compare the total sales between different seasons.
157 select season, sum(total_sales) as total_sales
158 from (
159     select
160         p.payment_value as total_sales,
161         case
162             when extract(month from o.order_purchase_timestamp) in (3, 4, 5) then 'Spring'
163             when extract(month from o.order_purchase_timestamp) in (6, 7, 8) then 'Summer'
164             when extract(month from o.order_purchase_timestamp) in (9, 10, 11) then 'Autumn'
165             else 'Winter'
166         end as season
167     from
168         amazon_brazil.orders o
169     join
170         amazon_brazil.payments p on o.order_id = p.order_id
171 ) as seasonal_sales
172 group by
173     season
174 order by
175     total_sales desc
176
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1

	season text	total_sales numeric
1	Spring	4934990.20
2	Summer	4890902.19
3	Winter	3416284.67
4	Autumn	2766695.06

Analysis: Sales are heavily influenced by seasonality, with clear peaks and troughs throughout the year. In Brazil, this corresponds to major holidays and events like Black Friday in Q4.

Recommendation for India: Create a granular marketing calendar based on India's diverse regional festivals (Onam, Durga Puja, Ganesh Chaturthi) in addition to national ones (Diwali, Holi, Eid). These cultural events are the primary drivers of consumer spending and present massive sales opportunities.

2. The inventory team is interested in identifying products that have sales volumes above the overall average.

SQL query snippet:

```
176
177 --3.2 The inventory team is interested in identifying products that have sales volumes above the overall a
178 with product_sales as (
179     select product_id, count(order_item_id) as total_quantity_sold
180     from amazon_brazil.order_items
181     group by product_id
182 )
183 select product_id, total_quantity_sold
184 from product_sales
185 where
186     total_quantity_sold > (select avg(total_quantity_sold) from product_sales)
187 order by total_quantity_sold desc
188 |
```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1

	product_id character varying (255)	total_quantity_sold bigint
1	aca2eb7d00ea1a7b8ebd4e6831466...	527
2	99a4788cb24856965c36a24e339b6...	488
3	422879e10f46682990de24d770e7f...	484
4	389d119b48cf3043d311335e499d9...	392
5	368c6c730842d78016ad823897a37...	388
6	53759a2ecddad2bb87a079a1f1519f...	373
7	d1c427060a0f73f6b889a5c7c61f2a	343

Total rows: 6366 Query complete 00:00:00.206

Analysis: This query pinpoints the "hero products" or "bestsellers" that consistently outperform the average item. These products have proven market fit and are highly popular with customers.

Recommendation for India: These products should be algorithmically promoted. Feature them on the homepage, place them higher in search results, and use them as anchors in "Frequently Bought Together" bundles. Ensure their supply chain is robust enough to handle sudden demand spikes.

3. To understand seasonal sales patterns, the finance team is analysing the monthly revenue trends over the past year (year 2018).

SQL query snippet:

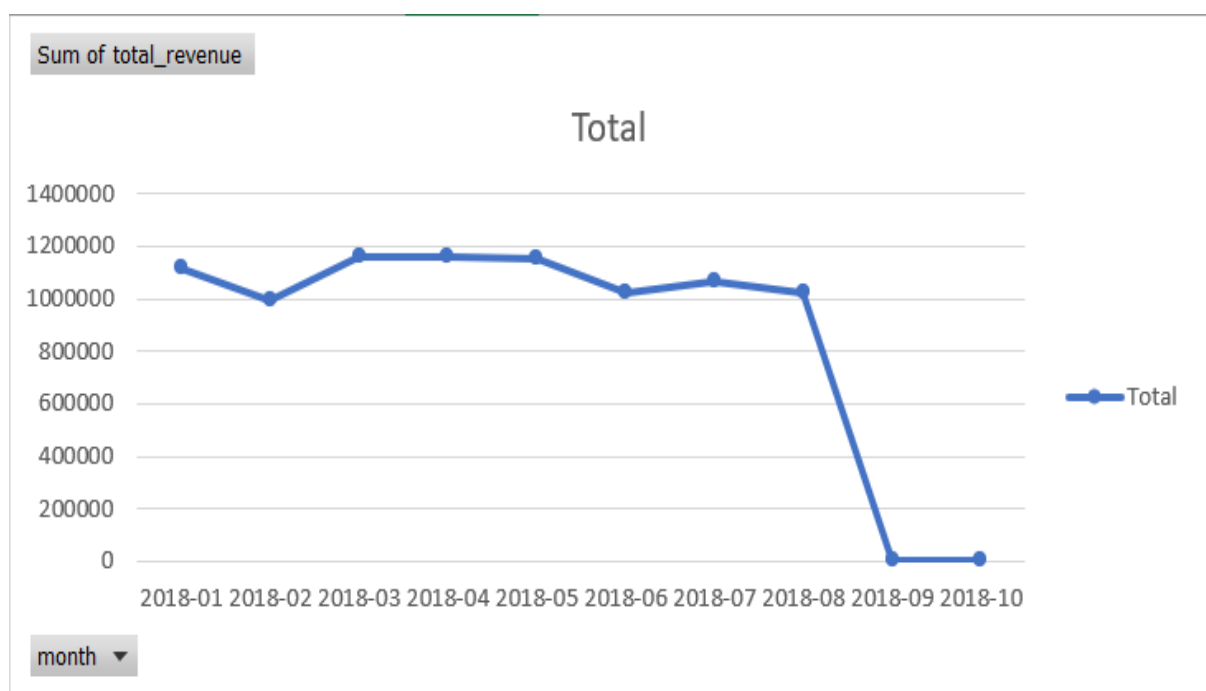
```
189
190 --3.3 To understand seasonal sales patterns, the finance team is analysing the monthly revenue trends over
191 select
192     to_char(o.order_purchase_timestamp, 'YYYY-MM') as month, sum(p.payment_value) as total_revenue
193 from amazon_brazil.orders o
194 join
195     amazon_brazil.payments p on o.order_id = p.order_id
196 where extract(year from o.order_purchase_timestamp) = 2018
197 group by month
198 order by month
199
```

	month text	total_revenue numeric
1	2018-01	1115004.18
2	2018-02	992463.34
3	2018-03	1159652.12
4	2018-04	1160785.48
5	2018-05	1153982.15
6	2018-06	1023880.50
7	2018-07	1066540.75
8	2018-08	1022425.32
9	2018-09	4439.54
10	2018-10	589.67

Analysis: The time-series data provides a clear visualization of the business's performance over the year. It highlights the overall growth trajectory and pinpoints specific months of peak performance versus periods of slower sales.

Recommendation for India: This monthly revenue chart should be a core component of the Amazon India business intelligence dashboard. It allows leadership to track performance against quarterly targets, measure the direct impact of marketing campaigns, and quickly identify any unexpected downturns that require immediate strategic intervention.

Row Labels ▾	Sum of total_revenue
2018-01	1115004.18
2018-02	992463.34
2018-03	1159652.12
2018-04	1160785.48
2018-05	1153982.15
2018-06	1023880.5
2018-07	1066540.75
2018-08	1022425.32
2018-09	4439.54
2018-10	589.67
Grand Total	8699763.05



4. A loyalty program is being designed for Amazon India.

SQL query snippet:

```
201 --3.4 A loyalty program is being designed for Amazon India.
202 with customer_order_counts as (
203     select c.customer_unique_id, count(o.order_id) as order_count
204     from amazon_brazil.customers c
205     join
206         amazon_brazil.orders o on c.customer_id = o.customer_id
207     group by c.customer_unique_id
208 ),
209 customer_segments as (
210     select
211         case
212             when order_count between 1 and 2 then 'Occasional'
213             when order_count between 3 and 5 then 'Regular'
214             when order_count > 5 then 'Loyal'
215         end as customer_type
216     from customer_order_counts
217 )
218 select customer_type, count(*) as count
219 from customer_segments
220 where customer_type is not null
221 group by customer_type
222 order by count DESC
223
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No:

	customer_type text	count bigint
1	Occasional	94635
2	Regular	333
3	Loyal	109

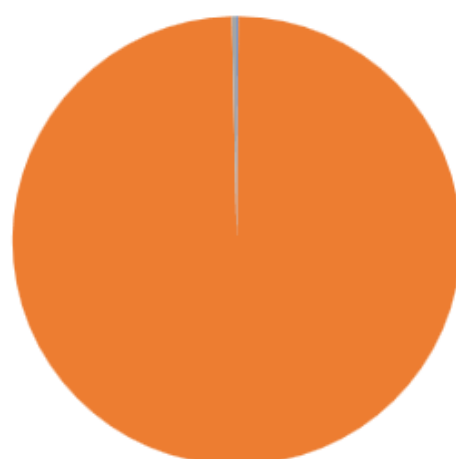
Analysis: This query provides a quantitative breakdown of the customer base by loyalty level ('Occasional', 'Regular', 'Loyal'). It answers a critical strategic question: "What proportion of our customers are casual shoppers versus deeply engaged, high-frequency buyers?"

Recommendation for India: Use this segmentation to design and evaluate the Amazon Prime program. The core business goal should be to create a value proposition (free shipping, content, exclusive deals) so compelling that it successfully converts 'Occasional' and 'Regular' shoppers into high-LTV (Lifetime Value) 'Loyal' Prime subscribers.

Row Labels	Sum of count
Loyal	109
Occasional	94635
Regular	333
Grand Total	95077

Sum of count

Total



customer_type ▼

- Loyal
- Occasional
- Regular

Chart Area

5. Amazon wants to identify high-value customers to target for an exclusive rewards program.

SQL query snippet:

```
225 --3.5 Amazon wants to identify high-value customers to target for an exclusive rewards program.
226 with order_values as (
227     select order_id, sum(payment_value) as total_order_value
228     from amazon_brazil.payments
229     group by order_id
230 ),
231 customer_avg_order_value as (
232     select o.customer_id, avg(ov.total_order_value) as avg_order_value
233     from amazon_brazil.orders o
234     join
235         order_values ov on o.order_id = ov.order_id
236     group by o.customer_id
237 )
238 select customer_id, avg_order_value,
239        rank() over (order by avg_order_value desc) as customer_rank
240 from customer_avg_order_value
241 limit 20
242
```

Data Output			Messages	Notifications
			SQL	Showing rows: 1 to 20 Page No:
	customer_id character varying (255)	avg_order_value numeric	customer_rank bigint	
1	1617b1357756262bfa56ab541c47b...	13664.0800000000000000	1	
2	ec5b2ba62e574342386871631fafd...	7274.8800000000000000	2	
3	c6e2731c5b391845f6800c97401a4...	6929.3100000000000000	3	
4	f48d464a0baaa338cb25f816991a...	6922.2100000000000000	4	
5	3fd6777bbce08a352fddd04e4a7cc...	6726.6600000000000000	5	
6	05455dfa7cd02f13d132aa7a6a972...	6081.5400000000000000	6	
7	df55c14d1476a9a3467f131269c24...	4950.3400000000000000	7	
8	e0a2412720e9ea4f26c1ac985f6a7...	4809.4400000000000000	8	
9	24bbf5fd2fe1b359ee7de94defc4a...	4764.3400000000000000	9	
Total rows: 20		Query complete 00:00:02.302		

Analysis: This query identifies a list of VIPs who spend the most money per transaction on average. These customers are extremely valuable due to the high margin on their orders, even if they don't purchase as frequently as others.

Recommendation for India: Create a "white-glove" service program for these high-value customers. This could include a dedicated (and human) customer support line, invitations to exclusive brand events, personalized thank-you notes, or early access to new product launches. Making these VIPs feel valued can foster immense loyalty.

6. Amazon wants to analyse sales growth trends for its key products over their lifecycle.

SQL query snippet:

```
243
244 --3.6 Amazon wants to analyze sales growth trends for its key products over their lifecycle.
245 with monthly_product_sales as (
246     select oi.product_id,
247           date_trunc('month', o.order_purchase_timestamp)::date as sale_month,
248           sum(oi.price) as monthly_sales
249     from amazon_brazil.order_items oi
250    join
251         amazon_brazil.orders o on oi.order_id = o.order_id
252    group by oi.product_id, sale_month
253 )
254 select product_id, sale_month,
255        sum(monthly_sales) over (partition by product_id order by sale_month) as total_sales
256 from monthly_product_sales
257 order by product_id, sale_month
258
```

	product_id character varying (255)	sale_month date	total_sales numeric
1	00066f42aeb9f3007548bb9d3f33c38	2018-05-01	101.65
2	00088930e925c41fd95ebfe695fd2655	2017-12-01	129.90
3	0009406fd7479715e4bef61dd91f2462	2017-12-01	229.00
4	000b8f95fcb9e0096488278317764d...	2018-08-01	117.80
5	000d9be29b5207b54e86aa1b1ac54...	2018-04-01	199.00
6	0011c512eb256aa0dbbb544d8dffcf6e	2017-12-01	52.00
7	00126f27c813603687e6ce486d909d...	2017-09-01	498.00
8	001795ec6f1b187d37335e1c470476...	2017-10-01	38.90
9	001795ec6f1b187d37335e1c470476...	2017-11-01	116.70

Total rows: 62176 Query complete 00:00:01.402

Analysis: The cumulative sales curve visualizes the commercial lifecycle of a product. A steep, upward-trending curve indicates a product in its high-growth phase. A curve that is beginning to flatten suggests the product is reaching market saturation or maturity.

Recommendation for India: Integrate this analysis into the inventory management system. For products with a steep growth curve, proactively increase purchase orders to avoid stockouts. For products with a flattening curve, consider running targeted promotions to clear remaining stock and plan for its replacement.

7. To understand how different payment methods affect monthly sales growth, Amazon wants to compute the total sales for each payment method and calculate the month-over-month growth rate for the past year (year 2018).

SQL query snippet:

```
260 --3.7 To understand how different payment methods affect monthly sales growth, Amazon wants to compute the
261 with monthly_payment_sales as (
262     select p.payment_type, DATE_TRUNC('month', o.order_purchase_timestamp)::date as sale_month,
263            sum(p.payment_value) as monthly_total
264     from amazon_brazil.payments p
265     join
266         amazon_brazil.orders o ON p.order_id = o.order_id
267     where extract(year from o.order_purchase_timestamp) = 2018
268     group by p.payment_type, sale_month
269 ),
270 sales_with_previous_month as (
271     select payment_type, sale_month, monthly_total,
272            lag(monthly_total, 1, 0) over (partition by payment_type order by sale_month) as previous_month_total
273     from monthly_payment_sales
274 )
275 select payment_type, sale_month, monthly_total,
276        case
277            when previous_month_total = 0 then null
278            else (monthly_total - previous_month_total) * 100.0 / previous_month_total
279        end as monthly_change
280 from sales_with_previous_month
281 order by payment_type, sale_month
```

Data Output Messages Notifications

Showing rows: 1 to 36 Page No: 1

	payment_type character varying (50)	sale_month date	monthly_total numeric	monthly_change numeric
1	boleto	2018-01-01	204844.66	[null]
2	boleto	2018-02-01	183112.72	-10.6089853648125365
3	boleto	2018-03-01	191538.02	4.6011549607258305
4	boleto	2018-04-01	193547.09	1.0489144661723035
5	boleto	2018-05-01	195378.93	0.94645700950605870644
6	boleto	2018-06-01	153350.28	-21.5113523244292514

Total rows: 36 Query complete 00:00:00.256

Analysis: This advanced query reveals the velocity of change in consumer payment preferences. It can show, for example, that a new digital wallet is rapidly gaining market share each month, while an older method like bank transfers is declining.

Recommendation for India: This is a mission-critical analysis for India's fast-evolving fintech landscape. If a new UPI feature or a specific bank's BNPL offering shows explosive month-over-month growth, Amazon India should move quickly to form a strategic partnership, offering co-branded promotions to align with emerging consumer habits and capture new customers.

THANK YOU!